GROUP AND RING THEORY - EXERCISE 12 (ANSWERS)

1. Obtain the Smith normal form and rank for the following matrices over a principal ideal domain R:

(a) $\begin{pmatrix} 0 & 2 & -1 \\ -3 & 8 & 3 \\ 2 & -4 & -1 \end{pmatrix}$, where $R = \mathbb{Z}$;

(b) $\begin{pmatrix} -x-3 & 2 & 0 \\ 1 & -x & 1 \\ 1 & -3 & -x-2 \end{pmatrix}$, where $R = \mathbb{Q}[x]$.

Solution. (a) We have

$$\begin{pmatrix} 0 & 2 & -1 \\ -3 & 8 & 3 \\ 2 & -4 & -1 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 10 \end{pmatrix}$$

so the rank is 3.

(b) Here

$$\begin{pmatrix} -x-3 & 2 & 0 \\ 1 & -x & 1 \\ 1 & -3 & -x-2 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & (x+1)^2(x+3) \end{pmatrix}.$$

The rank is then 3.

2. Find the invariant factors of the following matrix over $\mathbb{Q}[x]$:

$$\begin{pmatrix} 5-x & 1 & -2 & 4 \\ 0 & 5-x & 2 & 2 \\ 0 & 0 & 5-x & 3 \\ 0 & 0 & 0 & 4 \end{pmatrix}$$

Solution. We have

$$\begin{pmatrix} 5-x & 1 & -2 & 4 \\ 0 & 5-x & 2 & 2 \\ 0 & 0 & 5-x & 3 \\ 0 & 0 & 0 & 4 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & (5-x)^3 \end{pmatrix}$$

giving the invariant factors $1, 1, 1, (5-x)^3$.